PERIODICITY AND ANNUALIZED YIELDS

Learning Outcome

- calculate annual yield on a bond for varying compounding periods in a year

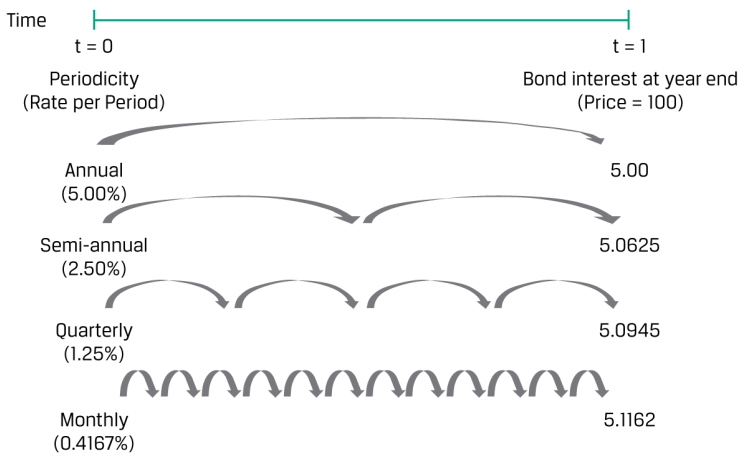
Investors analyzing bonds with various cash flow and maturity profiles seek a *standardized* yield measure to compare across different choices. Yield measures are usually *annualized* in order to allow a direct comparison. For capital market securities maturing in more than one year, investors want an *annualized* and *compounded* yield-to-maturity. Conventions for instruments maturing in one year or less will be covered in a later lesson.

An annualized and compounded yield on a fixed-rate bond depends on the assumed number of interest periods in the year, which is called the **periodicity** of the annual rate. The periodicity typically matches the frequency of coupon payments. For example, the fixed-coupon five-year bond issued by Bright Wheels Automotive (BRWA) from earlier lessons pays semiannual coupons with a stated annual yield-to-maturity of 3.2% for a periodicity of 2—the rate per semiannual period (1.6%) times 2. A bond that pays quarterly coupons has a stated annual yield for a periodicity of 4—the rate per quarter times 4.

Exhibit 1 shows how a 5% stated annual yield-to-maturity generates different amounts in one year with annual, semiannual, quarterly, monthly compounding based on a fixed-rate par bond priced at 100.

Exhibit 1:

Effective Annual Rate for Different Compounding Periods



The bond interest amounts on the right-hand side of Exhibit 1 expressed as percentages are known as effective annual rates. An **effective annual rate** has a periodicity of 1 because there is just one compounding period in the year. For example, for a quarterly coupon bond that pays 1.25% per period, or an annual coupon rate of (1.25% × 4) = 5%, each coupon received is assumed to be reinvested at the periodic rate—here, 1.25%—for the remainder of the year. We can solve for the future value of these cash flows using the Excel or Google Sheets FV function:

```
=FV(RATE, NPER, PMT, [PV], [TYPE]),
```

where

nper is the number of periods in a year

pmt is the rate per period

[*pv*] is the present value

[*type*] indicates when payments occur, with 0 designating payment at the end of each period

For the quarterly coupon bond:

=FV(0.0125,4,1.25,0,0) = 5.0945.

Therefore, assuming each coupon is reinvested at 1.25% through the end of the year, a quarterly coupon bond with an annual stated rate of 5% generates the same interest as an annual coupon bond with a stated rate of 5.0945%. The Excel or Google Sheets EFFECT function may also be used to calculate the effective annual rate for a given number of compounding periods per year.

The most common periodicity in the United States, the United Kingdom, and several other fixed-income markets is 2 because most bonds in these jurisdictions make semiannual coupon payments. An annual rate having a periodicity of two is known as a **semiannual bond basis yield**, or **semiannual bond equivalent yield**. Therefore, a semiannual bond basis yield is the yield per semiannual period times 2. It is important to remember that "semiannual bond basis yield" and "yield per semiannual period" have different meanings. For example, if a bond yield is 2% per semiannual period, its annual yield is 4% when stated on a semiannual bond basis.

EXAMPLE 1

Periodicity of Zero-Coupon Bonds

The periodicity of the annual market discount rate for a zero-coupon bond is arbitrary because there are no coupon payments. Consider a five-year, zero-coupon bond priced at 80 per 100 of par value. Assuming annual compounding, or a periodicity of one, the yield-to-maturity is 4.564%.

$$80 = \frac{100}{(1+r)^5}; r = 0.045640; 4.564\%$$

Assuming quarterly compounding, or a periodicity of four, the annual yield-to-maturity for this same five-year, zero-coupon bond is 4.488%.

$$80 = \frac{100}{(1+r)^{20}}; r = 0.011220; \times 4 = 0.044880.$$

As shown in Exhibit 2, 2.2565% compounded 2 times a year, 1.1220% compounded 4 times a year, and 0.3726% compounded 12 times a year are all equivalent to an effective annual rate of 4.564%.

Exhibit 2:

Periodicity of Annual Discount Rate for Zero-Coupon Bond

Column 1	*Column 2*	*Column 3: (Col. 2 × 5)*	*Column 4*	*(Col. 2 × Col. 4)*	*Column 6: (1+Col. 4)^(Col. 2)*
Periodicity	**Number of Periods/ Year**	**Number of Compounding Periods**	**Periodic Yield-to-Maturity (%)**	**Annual Yield- to-Maturity (%)**	**Periodic YTM Compounded at Periods/ Year (%)**
Annual	1	5	0.045640	0.045640	0.045640
Semi-annual	2	10	0.022565	0.045130	0.045640
Quarterly	4	20	0.011220	0.044879	0.045640
Monthly	12	60	0.003726	0.044712	0.045640

The compounded total return is the same for each expression for the annual rate. They differ in terms of the number of compounding periods per year—that is, in terms of the *periodicity* of the annual rate. For a given pair of cash flows, the stated annual rate and the periodicity are inversely related.

An important tool in fixed-income analysis is the conversion of an annualized yield using one periodicity to another periodicity. These are called periodicity, or compounding, conversions. Equation 1 shows a general formula to convert an annual percentage rate (APR) for m periods per year, denoted APR_m, to an annual percentage rate for n periods per year, APR_n:

$$\left(1+\frac{APR_m}{m}\right)^m=\left(1+\frac{APR_n}{n}\right)^n. \qquad 1$$

For example, suppose that a three-year, 5% semiannual coupon payment corporate bond is priced at 104 per 100 of par value. Its yield-to-maturity is 3.582%, quoted on a semiannual bond basis for a periodicity of 2: 0.01791 × 2 = 0.03582.

$$104=\frac{2.5}{(1+r)^1}+\frac{2.5}{(1+r)^2}+\frac{2.5}{(1+r)^3}+\frac{2.5}{(1+r)^4}+\frac{2.5}{(1+r)^5}+\frac{102.5}{(1+r)^6};\ r=0.01791.$$

To compare this bond with others, an analyst converts this annualized yield-to-maturity to quarterly and monthly compounding. Doing so entails using Equation 1 to convert from a periodicity of m = 2 to periodicities of n = 4 and n = 12.

$$\left(1+\frac{0.03582}{2}\right)^2=\left(1+\frac{APR_4}{4}\right)^4;\ APR_4=0.03566.$$

$$\left(1+\frac{0.03582}{2}\right)^2=\left(1+\frac{APR_{12}}{12}\right)^{12};\ APR_{12}=0.03556.$$

An annual yield-to-maturity of 3.582% for semiannual compounding provides the same rate of return as annual yields of 3.566% and 3.556% for quarterly and monthly compounding, respectively. A general rule for periodicity conversions is *compounding more frequently at a lower annual rate corresponds to compounding less frequently at a higher annual rate*. This rule can be used to check periodicity conversion calculations.

EXAMPLE 2

Comparing Strip Bonds to Coupon Bonds

Financial intermediaries can own coupon bonds and then issue individual zero-coupon bonds composed of coupon or principal payments from the coupon bonds. These zero-coupon bonds are known as strip bonds. Strip bonds are attractive to investors seeking to match known cash flows on a specific future date.

A 20-year government of Canada strip (zero-coupon) bond is priced at 69.4300 per 100. An analyst wants to compare the yield of the strip bond with that of a newly issued 20-year coupon bond. Both the strip bond and the coupon bond are non-callable. The 20-year coupon bond pays semiannual coupons of 2%, and its current

The analyst will compute a semiannual bond equivalent yield for the strip bond and compare it to the semiannual bond equivalent yield of 1.880% of the 20-year coupon bond.

There are two approaches. The first involves calculating a yield-to-maturity assuming annual compounding for the strip bond and converting it to a semiannual equivalent bond yield. The second approach involves directly computing a semiannual yield-to-maturity for the strip bond and annualizing it by multiplying it by 2.

Approach 1

The 20-year strip bond does not pay any coupons, so as the first step, we assume annual compounding (periodicity of 1) to calculate an effective annual rate based on the current price and remaining years to maturity.

$$69.43 = \frac{100}{(1+r)^{20}}; r = 0.01840997 = 1.841\%.$$

The next step is to convert the effective annual rate to semiannual bond equivalent yield. This is done by using Equation 1 to convert from a periodicity of 1 to a periodicity of 2.

$$\left(1 + \frac{APR_m}{m}\right)^m = \left(1 + \frac{APR_n}{n}\right)^n.$$
$$\left(1 + \frac{0.01841}{1}\right)^1 = \left(1 + \frac{APR_2}{2}\right)^2.$$
$$APR_2 = 0.01832604 = 1.833\%.$$

Approach 2

Alternatively, we can assume that compounding happens semiannually (periodicity of two) and the effective rate calculated would be semiannual.

$$69.43 = \frac{100}{(1+r)^{40}}; r = 0.009163 = 0.916\%.$$

To annualize this, we multiply it by 2 and obtain the same result as the first approach: 1.833%.

EXAMPLE 3

Periodicity Conversion with Negative Yields

Assume the German government issues a five-year zero-coupon bond at a price of 103.72. Calculate the effective annual rate of the bond at issuance assuming annual, semiannual, and monthly compounding.

Hide Solution

Solution:

First, we can calculate the German bond's yield-to-maturity as an effective annual rate assuming the periodicity of 1.

$$103.72 = \frac{100}{(1+r)^5}; r = -0.00727834 = -0.7278\%.$$

Converting that yield to semiannual and monthly compounding requires using Equation 1 to convert to from a periodicity of 1 to periodicities of 2 and 12, respectively.

Conversion of an annual yield (m = 1) to semiannual (n = 2):

$$\left(1+\frac{APR_2}{1}\right)^1 = \left(1+\frac{APR_2}{2}\right)^2.$$
$$APR_2 = -0.007291 = -0.7291\%.$$

Conversion of an annual yield (m = 1) to monthly (n = 12):

$$\left(1+\frac{APR_m}{m}\right)^m = \left(1+\frac{APR_n}{n}\right)^n.$$
$$\left(1+\frac{-0.007278}{1}\right)^1 = \left(1+\frac{APR_{12}}{12}\right)^{12}.$$
$$APR_2 = -0.007303 = -0.7303\%.$$

Note: Quantities in the calculation are not rounded, though they are shown with three decimal places for presentation purposes. Solutions are rounded to three decimal places.

Question Set

1. A bond has a yield-to-maturity of –0.50% using annual compounding. If the yield is converted to monthly compounding, it will *most likely* be:

A. greater than –0.50%.
B. equal to –0.50%.
C. less than –0.50%.

Hide Solution

Solution:

C is correct. A general rule for periodicity conversions is compounding more frequently within the year corresponds to a lower (or more negative) yield-to-maturity.

2. Convert a yield of 1.736% compounded monthly to an effective annual rate.

Hide Solution

Solution:

The effective annual rate assumes only one compounding period in a year (periodicity of 1). Converting from monthly to annual compounding requires converting from a periodicity of 12 to 1:

$$\left(1+\frac{APR_m}{m}\right)^m = \left(1+\frac{APR_n}{n}\right)^n.$$
$$\left(1+\frac{0.01736}{12}\right)^{12} = \left(1+\frac{APR_1}{1}\right)^1.$$
$$APR_1 = 0.0175 = 1.75\%.$$

3. When calculating yield-to-maturity for a strip (zero-coupon) bond, the periodicity must be 1.

Justify your answer.

A. True
B. False

Hide Solution

Solution

False. The periodicity of the annual market discount rate for a zero-coupon bond is arbitrary because there are no coupon payments. For example, a yield

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If anyone is curios how to do example 1 on a calculator.

First, set to quarterly compounding:

2nd + P/Y set to 4 + Enter

2nd + Quit

Now you can solve for the 0-coupon quarterly compounded bond:

N = 5 * 4 = 20

PV = -80

PMT = 0

FV = 100

CPT I/Y = 4.48786%

Created 3 months ago by Maxim Ovchinnikov

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IS

For question 1) shouldn't the right answer be A)? Since the monthly compounding figure is -0.4997% (greater than -0.5%)

Created 5 months ago by Imre Szephalmi

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